Grendel Taishi Gardiner

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Technical Summary

Mechanical Engineer pursuing a BS/MS at Georgia Tech focused on robotic manipulation, physically interactive robotic systems, and design for robotic and medical devices.

Education | Georgia Institute of Technology | Atlanta, GA, USA

Master of Science in Mechanical Engineering — *Expected* May 2027
Bachelor of Science in Mechanical Engineering, GPA: 3.6/4.0 — May 2026

Research and Engineering

BM2 Lab – Modular self-reconfigurable Continuum Robot | *Undergraduate Research Assistant* — Aug 2025 – Present

- Optimized stability using stiffness modeling and FEA, developing a lookup table reducing joint selection time by 50%.
- Developed simulation-driven continuum robot workflow using MuJoCo, MATLAB, and FEA to evaluate tendon-driven manipulation and constrained-environment interaction.

Tendonova, Atlanta, GA | *R&D Intern* — Jan 2025 – Present

- Designed and prototyped minimally invasive end-effectors for hard-tissue debridement operations, incorporating injection molding constraints, tolerance analysis, and rapid iteration workflows.
- Redesigned luer locking mechanism based on customer feedback to improve usability and reduce complications during operation; worked within FDA 21 CFR Part 820 QMS, collaborating with clients and manufacturer to refine design.
- Developed Hall-effect-sensor-based closed-loop oscillation control system for compact device actuation, regulating tool-head frequency within ±5% of target operation.

GT Medical Robotics Club | *Project Lead – Tendon-Driven Anthropomorphic Robotic Thumb* — Jan 2025 – Present

- Designed tendon-driven anthropomorphic thumb mechanism with compact internal actuation for dexterous manipulation and compliant interaction.
- Evaluated actuator/transmission trade-offs (cycloidal, harmonic, and planetary), balancing torque and compactness.
- Developing wearable sensing and MuJoCo workflows for closed-loop intent-driven manipulation.

Bhamla Lab – Elastic Kink Deformation of Soft Metamaterial | *Undergraduate Research Assistant* — Aug 2024 – Aug 2025

- Conducted nonlinear FEA and analytical modeling of chiral buckling meta-structures to characterize geometric parameter effects on energy storage and compliant mechanical behavior.
- Orally presented elastic metamaterial jumping capabilities at the 2025 *American Physics Society* conference.

Kickr Design, Atlanta, GA | *Engineering Intern* — Aug 2024 – Dec 2024

- Designed rapid-prototyping fixtures, manufacturing jigs, and snap-fit mechanisms to improve fabrication efficiency and assembly repeatability.

Robotics Projects

Bio-Inspired Tree Climbing Quadruped — Jan 2026 – Present

- Designed climbing quadruped with compliant universal-joint ankles and microspine adhesion mechanisms for stable contact on irregular tree surfaces..
- Developed MuJoCo-based locomotion pipeline using PPO reinforcement learning and sim-to-real gait validation for autonomous crawling and terrain interaction.
- Investigating computer-vision-based bark inspection and adaptive locomotion planning.

Dynamic Hoop Stabilizing Robot — Jan 2026 – May 2026

- Modeled and controlled nonlinear hoop dynamics by regulating motor-driven normal force against gravity and friction using ToF-based feedback, Kalman-filter state estimation, and lead-lag control.

Skills

Robotics & Controls: Kinematics, PID Control, PPO RL, Tendon-Driven Systems, Continuum Robotics, Sim-to-Real Validation
Simulation & Analysis: ANSYS Workbench, MuJoCo, MATLAB, FEA, Stiffness Modeling
Design and Fabrication: Onshape (5+ years), SolidWorks, Injection Molding, CNC Machining, SLS/FDM Printing, Rapid Prototyping
Embedded Systems: PCB Design, Hall Effect Sensing, PWM Control, Arduino, STM32
Languages: English (native) Japanese (native)